AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Science and Technology (FST)
Department of Computer Science (CS)
Undergraduate Program

COURSE PLAN — SEMESTER: Spring 2025-2026

I. Course Core and Title

CSC 2106: Data Structure (Theory)

II. Credit

3 credit hours (3 hours of theory per week)

III. Nature

Core Course for CSE

IV. Prerequisite

CSC 1204: Discrete Mathematics
CSC 1205 Object Oriented Programming

V. Vision:
Our vision is to be the preeminent Department of Computer Science through creating recognized professionals who will provide innovative solutions by leveraging contemporary research methods and development techniques of computing that is in line with the national and global context.

VI. Mission:
The mission of the Department of Computer Science of AIUB is to educate students in a student-centric dynamic learning environment; to provide advanced facilities for conducting innovative research and development to meet the challenges of the modern era of computing, and to motivate them towards a life-long learning process.

VII - Course Description:

- Discuss class policy, course content, different data types, introduction to data structure.
- Explain Arrays, linear searching, insertion, and deletion operations on array.
- Parallel and Multidimensional array Iteration
- ADT Data type and recursion
- Stack operation [initialization, push, pop, Isempty]
- Implementation with array
- Arithmetic Expression conversion (Infix, Postfix, Prefix) and Expression evaluation
- Linear & Circular queue operation [initialization, enqueue, dequeue, Isempty]
- Bubble sort, Insertion sort and Selection sort algorithms and Binary search algorithms.
- Complexity Analysis (Space and Time)
- Time Complexity with respect to Array, Pointer, and Structure and for all the next topics and make comparison if possible.
- Understand Dynamic storage management
- Singly Linked List
- Insertion, Deletion and Search Operation of Singly Linked List
- Binary, complete, full, and balanced tree,
- Tree traversal-inorder, preorder, postorder
- Construction of binary search tree, Searching, Insertion and Deletion operation with simulations. Representation: adjacency matrix & List, spanning tree, Traversing method (BFS and DFS)

VIII – Course outcomes (CO) Matrix:

By the end of this course, students should be able to:

COs *	Description	Domain Level ***			PO Assessed ****
		C	P	A	

CO1 **	**Understand** different concepts of data storage and retrieve procedures in computer memory	3			PO-a-1
CO2	**Demonstrate** the selective well data structure techniques to solve some engineering problems.	3			PO-a-2
CO3	**Determine** the usage of different data structures to solve some complex realistic problems.	3			PO-a-2
CO4 **	**Justify** data structures concepts in designing and developing solution for problems.	3			PO-b-3

C: Cognitive; P: Psychomotor; A: Affective Domain

* *CO assessment method and rubric of COs assessment is provided in later section*
** *COs will be mapped with the Program Outcomes (POs) for PO attainment*
*** *The numbers under the 'Level of Domain' columns represent the level of Bloom's Taxonomy each CO corresponds to.*
**** *The numbers under 'PO Assessed' column represent the POs each CO corresponds to.*

IX – Topics to be covered in the class:*

Time Frame	CO Mapped	Topics	Teaching Activities	Assessment Strategy(s)
Week 1	CO1, CO2	Mission and Vision, Elementary data types and introduction to data structures, 1D Array	Lecture, Question-answer	Quiz, Term Exam
Week 2	CO1, CO2	2D Array and String	Lecture, Question-answer	Quiz, Term Exam
Week 3	CO1, CO2	Structure	Lecture, Question-answer	Quiz, Term Exam
Week 4	CO1, CO4	Sorting: Bubble, Selection, Insertion Sort	Lecture, Question-answer	Quiz, Term Exam
Week 5	CO1, CO4	Searching: Linear & Binary Search	Lecture, Question-answer	Quiz, Term Exam
Week 6	CO1, CO2	Stack and Stack Applications	Lecture, Question-answer	Quiz, Term Exam
Midterm (Week 7)				
Week 8	CO1, CO3	Queue Queue Applications	Lecture, Question-answer	Quiz, Term Exam
Week 9	CO1, CO3	Pointer, Linked List: Singly	Lecture, Question-answer	Quiz, Term Exam
Week 10	CO3, CO4	Linked List: Doubly	Lecture, Question-answer	Quiz, Term Exam
Week 11	CO3, CO4	Tree and Tree Applications	Lecture, Question-answer	Quiz, Term Exam
Week 12	CO3, CO4	BST & Heap	Lecture, Question-answer	Quiz, Term Exam
Week 13	CO3, CO4	Graph, Graph representation, BFS, DFS	Lecture, Question-answer	Quiz, Term Exam
Final term (Week 14)				

** The faculty reserves the right to change, amend, add, or delete any of the contents.*

X – Mapping of PO/PLO and K, P, A of this course:

PO Indicator ID	PO Indicators Definition (As per the requirement of WKs)	Domain	K	P	A
PO-a-1	Apply information and concepts in natural science with the familiarity of issues.	Cognitive Level 3 (Applying)	K1		
PO-a-2	Apply information and concepts of mathematics with the familiarity of issues.	Cognitive Level 3 (Applying)	K2		
PO-b-3	Analyze solutions for complex engineering problems reaching substantiated conclusion.	Cognitive Level 5 (Evaluating)	K3	P1 P3	

XI – K, P, A Definitions

Indicator	Title	Description
K1	Theory based natural science	A systematic, theory-based understanding of the natural sciences applicable to the discipline
K2	Conceptual based mathematics	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline
K3	Theory based engineering fundamentals	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
P1	Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach
P3	Depth of analysis required	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models

XII – Mapping of CO Assessment Method and Rubric

The mapping between Course Outcome(s) (COs) and The Selected Assessment method(s) and the mapping between Assessment method(s) and Evaluation Rubric(s) is shown below:

COs	Description	Mapped POs	Assessment Method	Assessment Rubric
CO1	**Understand** different concepts of data storage and retrieve procedures in computer memory	PO-a-1	Quiz	Rubric for Quiz
CO2	**Demonstrate** the selective well data structure techniques to solve some engineering problems.	PO-a-2	Quiz	Rubric for Quiz
CO3	**Determine** the usage of different data structures to solve some complex realistic problems.	PO-a-2	Quiz	Rubric for Quiz

CO4	**Justify** data structures concepts in designing and developing solution for problems	PO-b-3	Term Exam	Rubric for Term Exam

XIII – Evaluation and Assessment Criteria

CO1: Understand different concepts of data storage and retrieve procedures in computer memory

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				
Data storage	Explaining suitable method to derive complexity of the given algorithms				
Data retrieve	Derive mathematical function using the appropriate method in the first step.				
Type of Data Structures	Use standard complexity notations like Big O, Big Ω and Big Θ to compute complexity.				

CO2: Demonstrate the selective well data structure techniques to solve some engineering problems.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				
Problem Understand	Understand the problem context and nature of the problem. Describe appropriate data structure to analyze the problem.				
Selection of Data Structure	Linking the type of data structure available in order to solve engineering problems				
Demonstrate the technique	Apply suitable techniques for the given problem to solve and demonstrate the process.				

CO3: Determine the usage of different data structures to solve some complex realistic problems.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				
Problem Analysis	Understand the problem context and nature of the problem. Describe appropriate data structure technique to analyze the problem.				
Use of proper data structure	Apply the appropriate data structure method to solve the problem and demonstrate the process.				
Efficiency	Describe details of the proposed technique with performance efficiency.				

CO4: Justify data structures concepts in designing and developing solution for problems

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				
Problem Analysis	Understand the problem context and nature of the problem. Describe appropriate data structure approach to analyze the problem.				
Algorithm Selection based on appropriate choosing of data structure	Explore different dimensions of problems considering different type of data structures and choose the correct algorithm to solve the problem.				
Justification of the solution	Justify the solution after applying appropriate data structure on that problem.				

XIV- Course Requirements

- Students are expected to attend at least 80% of the class.
- Students are expected to participate actively in the class.
- For both terms, there will be at least 2 quizzes based on the theoretical knowledge and conceptual understanding of the topic covered discussed in the classes.
- Submit report based on the given course related problems.
- Submission of assignment and projects should be in due time.

XV – Evaluation & Grading System*

The following grading system will be strictly followed in this class

Mid Term Exam:
Term Exam: 60%
Quizzes: 30%
Attendance & Performance: 10%
Final Term Exam:
Term Exam: 60%
Quizzes: 30%
Attendance & Performance: 10%

Semester grade: 40% midterm + 60% final term

Letter	Grade Point	Numerical %
A+	4.00	90-100
A	3.75	85 - < 90
B+	3.50	80 - < 85
B	3.25	75 - < 80
C+	3.00	70 - < 75
C	2.75	65 - < 70
D+	2.50	60 - < 65
D	2.25	50 - < 60
F	0.00	< 50
I		Incomplete
W		Withdrawal
UW		Unofficially Withdrawal

** The evaluation system will be strictly followed as par the AIUB grading policy.*

XVI – Textbook/ References

- **"Schaum's Outline of Data Structures with C++"**. By John R. Hubbard
- **"Data Structures and Program Design",** Robert L. Kruse, 3rd Edition, 1996.
- **"Data structures, algorithms and performance",** D. Wood, Addison-Wesley, 1993
- **"Advanced Data Structures",** Peter Brass, Cambridge University Press, 2008
- **"Data Structures and Algorithm Analysis",** Edition 3.2 (C++ Version), Clifford A. Shaffer, Virginia Tech, Blacksburg, VA 24061 January 2, 2012
- **"C++ Data Structures",** Nell Dale and David Teague, Jones and Bartlett Publishers, 2001.
- **"Data Structures and Algorithms with Object-Oriented Design Patterns in C++",** Bruno R. Preiss,

XVII - List of Faculties Teaching the Course

FACULTY NAME	SIGNATURE
ANEEM AL AHSAN RUPAI	
DIPTA JUSTIN GOMES	
JUENA AHMED NOSHIN	
KAZI TANVIIR	
MAHFUJUR RAHMAN	
MD. HAMID UDDIN	
MD. HASIBUL HASAN	
MD. RAIHAN MAHMUD	
NYME AHMED	
PROF. DR. MD. RAFIQUL ISLAM	
S M ABDULLAH SHAFI	
SHAIKAT DAS JOY	
SYEDA ANIKA TASNIM	
SYMA KAMAL CHAITY	

XVIII – Verification:

<table>
<tr><td colspan="2">Prepared by:

SHAIKAT DAS JOY
Course Convener

Date:.....</td><td colspan="2">Moderated by:

Dr. M. Mahmudul Hasan
Point Of Contact
OBE Implementation Committee for CS

Date:.....</td></tr>
<tr><td>Checked by:

.....

Dr. Muhammad Firoz Mridha
Head (Undergraduate Program)
Department of Computer Science

Date:.....</td><td>Certified by:

.....

Dr Md Abdullah Al Jubair
Director,
Faculty of Science & Information Technology

Date:.....</td><td colspan="2">Approved by:

.....
Dr. Dip Nandi
Associate Dean,
Faculty of Science & Information Technology

Date:.....</td></tr>
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